

THE FIFTH WILLIAM LOWELL PUTNAM MATHEMATICAL COMPETITION

March 7, 1942

Morning Session

1. A square of side $2a$, lying always in the first quadrant of the XY plane, moves so that two consecutive vertices are always on the X - and Y -axes respectively. Find the locus of the midpoint of the square. (page 184)

2. If a polynomial $f(x)$ is divided by $(x - a)^2 (x - b)$, where $a \neq b$, derive a formula for the remainder. (page 185)

3. Is the following series convergent or divergent?

$$1 + \frac{1}{2} \cdot \frac{19}{7} + \frac{2!}{3^2} \left(\frac{19}{7}\right)^2 + \frac{3!}{4^3} \left(\frac{19}{7}\right)^3 + \frac{4!}{5^4} \left(\frac{19}{7}\right)^4 + \dots$$

(page 186)

4. Find the orthogonal trajectories of the family of conics $(x + 2y)^2 = a(x + y)$. At what angle do the curves of one family cut the curves of the other family at the origin? (page 187)

5. A circle of radius a is revolved through 180° about a line in its plane, distant b from the center of the circle, where $b > a$. For what value of the ratio b/a does the center of gravity of the solid thus generated lie on the surface of the solid? (page 189)

6. Any circle in the XY (horizontal) plane is "represented" by a point on the vertical line through the center of the circle and at a distance "above" the plane of the circle equal to the radius of the circle.

Show that the locus of the representations of all the circles which cut a fixed circle at a constant angle is a (portion of a) one-sheeted hyperboloid.

By consideration of suitable families of circles in the plane, demonstrate the existence of two families of rulings on the hyperboloid. (page 191)

Afternoon Session

7. A square of side $2a$, lying always in the first quadrant of the XY plane, moves so that two consecutive vertices are always on the X - and Y -axes respectively. Prove that a point within or on the boundary of the square will

in general describe a (portion of a) conic. For what points of the square does this locus degenerate? (page 194)

8. For the family of parabolas

$$y = \frac{a^3 x^2}{3} + \frac{a^2 x}{2} - 2a,$$

- (i) find the locus of vertices,
- (ii) find the envelope,
- (iii) sketch the envelope and two typical curves of the family. (page 195)

9. Given

$$x = \phi(u, v)$$

$$y = \psi(u, v)$$

where ϕ and ψ are solutions of the partial differential equation

$$(1) \quad \frac{\partial \phi}{\partial u} \frac{\partial \psi}{\partial v} - \frac{\partial \phi}{\partial v} \frac{\partial \psi}{\partial u} = 1.$$

By assuming that x and v are the independent variables, show that (1) may be transformed to

$$(2) \quad \frac{\partial y}{\partial v} = \frac{\partial u}{\partial x}.$$

Integrate (2), and show how this effects in general the solution of (1). What other solutions does (1) possess? (page 198)

10. A particle moves under a central force inversely proportional to the k th power of the distance. If the particle describes a circle (the central force proceeding from a point *on the circumference of the circle*), find k .

(page 201)

11. Sketch the curve

$$y = \frac{x}{1 + x^6 \sin^2 x},$$

and show that

$$\int_0^\infty \frac{x \, dx}{1 + x^6 \sin^2 x}$$

exists.

(page 203)